



اُنِيُوَرْسِيْتِي تِيكْنُوْلُوْجِي مَارَا
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TEKNOLOGI
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REPORT

E-COMMUNITY SERVICE & VOLUNTEER PLATFORM

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1.0 Introduction

The E-Community Service & Volunteer Platform is a digital solution designed to connect citizens with meaningful volunteer programs and social work opportunities. The platform serves as a centralized hub where users can easily discover, register and participate in community service initiatives organized by various organizations and institutions.

Through this platform, citizens can fulfill community service requirements while gaining valuable real-world experience and citizens can actively contribute to social causes that align with their interests. By simplifying the volunteer registration process and improving communication between organizers and participants, the platform promotes greater community engagement, social responsibility and collaborative efforts to address societal needs.

1.1 Objectives

The E-Community Service & Volunteer Platform is a web-based system designed to connect citizens with opportunities to engage in volunteer programs and social activities. The platform provides a structured and accessible environment where individuals can explore ways to contribute to their communities, develop a sense of social responsibility and participate in meaningful initiatives.

This system aims to enhance community engagement and promote a culture of volunteerism. By offering a reliable and organized platform, it encourages individuals to take part in social programs and activities that support the development of their communities. In addition, the platform seeks to improve the organization and coordination of community service efforts to ensure that initiatives are planned and managed in an effective and transparent manner.

Overall, the E-Community Service & Volunteer Platform supports the growth of a socially responsible and caring society. It motivates people to actively contribute to social causes, fostering empathy, collaboration and awareness of societal needs. By doing so, the platform aims to create long-term benefits for both individuals and communities to strengthen the bonds within society and promote positive social change.

2.0 Problem Statement

There are three critical issues hindered in the local community despite the willingness of many citizens to contribute

1. Fragmentation of information

There is lack of a centralized platform, making it difficult for individuals to discover and access volunteer opportunities that match their skills and interest

2. Operational inefficiency

Volunteer coordination and registration remain heavily reliant on manual processes leading to administrative bottlenecks and poor communication between organizer and participants

3. Low engagement level

General awareness regarding community needs is insufficient resulting in low participation rates and a missed opportunity to build a culture of social responsibility

3.0 Database Design

E-Community Platform

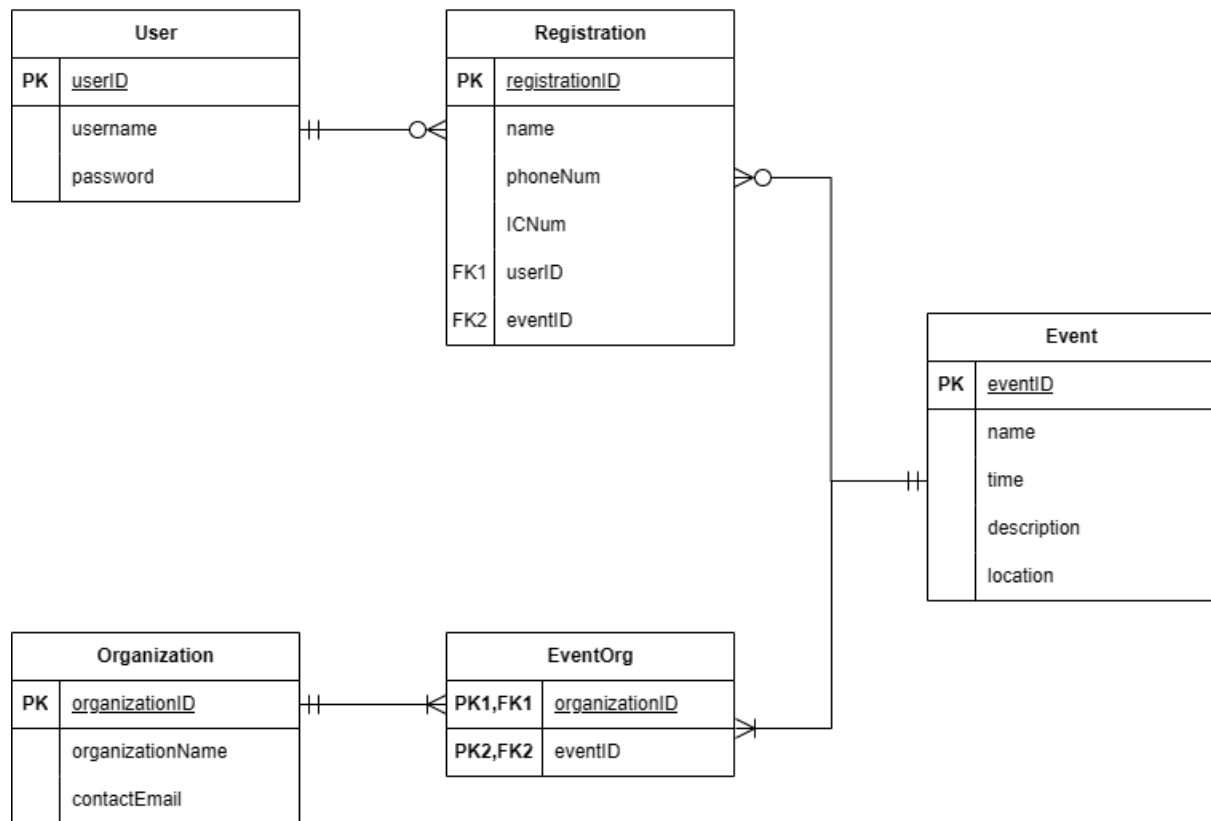


Figure 1.0 shows the ERD for the E-Community Service and Volunteer Platform

Our ERD consists of 5 tables which are User, Registration, Event, EventOrg and Organization. A user can register for many events and an event can have many users registered. The registration entity represents this many-to-many relationship between user and event. Each registration belongs to only one user and one event, which is why userID and eventID are included as foreign keys in the Registration table.

An organization can organize many events and an event can be organized by many organizations. The EventOrg entity acts as a bridge table to represent this many-to-many relationship, and therefore contains organizationID and eventID as foreign keys.

4.0 MVC Framework Design

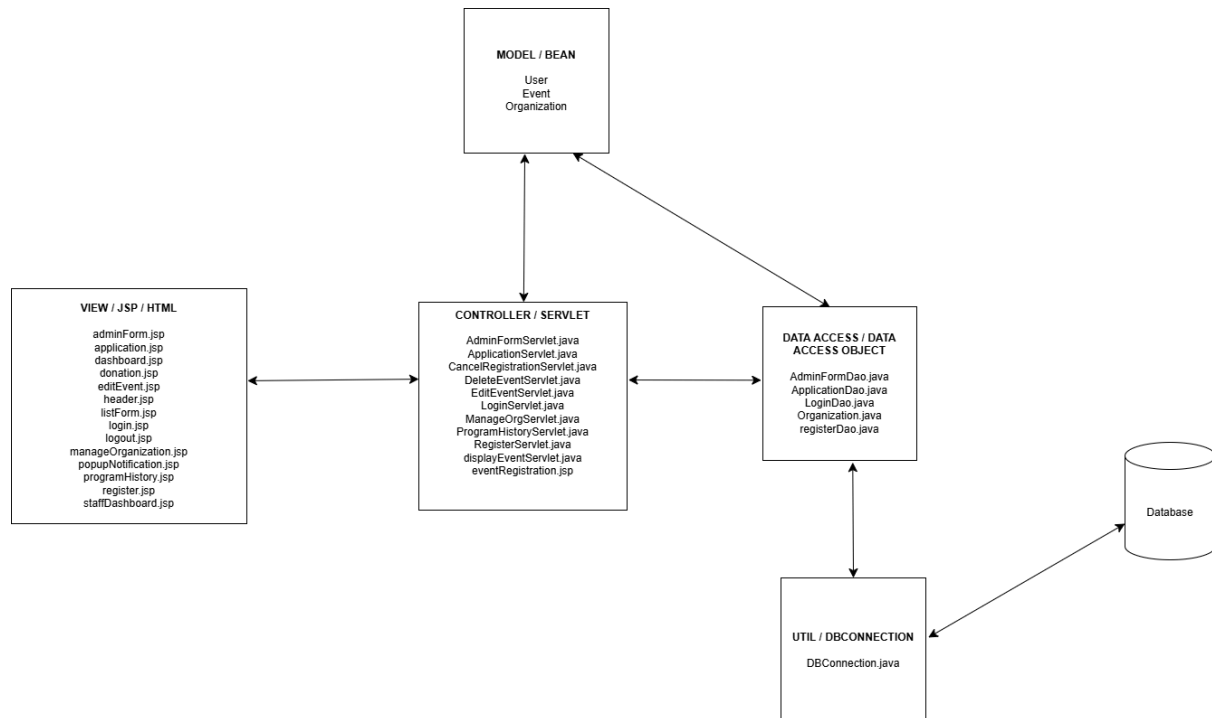


Figure 4.0 shows the Model View Controller (MVC) Framework Design for the E-Community Service and Volunteer Platform

Our system platform is built using Model View Controller (MVC) architecture with an additional Data Access Object (DAO) layer. This architecture creates a clear, unidirectional flow of data through the system.

The view layer consists of JSP pages that display data to users and capture their inputs. The controller layer consists of servlets which receive all requests from view and validate the input, implements business logic and coordinates between view and data layers. The model layer contains java objects serving as data containers with getter/setter methods. The dao layer handles all database operations by executing SQL queries and converting results into model objects. The dbconnection layer provides database connection and centralizes database credentials. Finally the database layer that stores all data and executes queries sent from the dao layer.

USER MANUAL

E-Community Service and Volunteer Platform

5.0 User Manual

1. Homepage

Figure 1.0 , figure 1.1 and figure 1.2 illustrate a homepage of the e-community service and volunteer platform which clearly states the organization's core business including the vision and mission. Users can have the insight of the organization by just viewing this homepage and can easily understand the business operations. The organization also welcomes a donor to donate for this organization as this organization is a volunteer based organization, so it does not have a big financial system. With the donation, the organization can hold many more volunteer programs with the help of financial support.

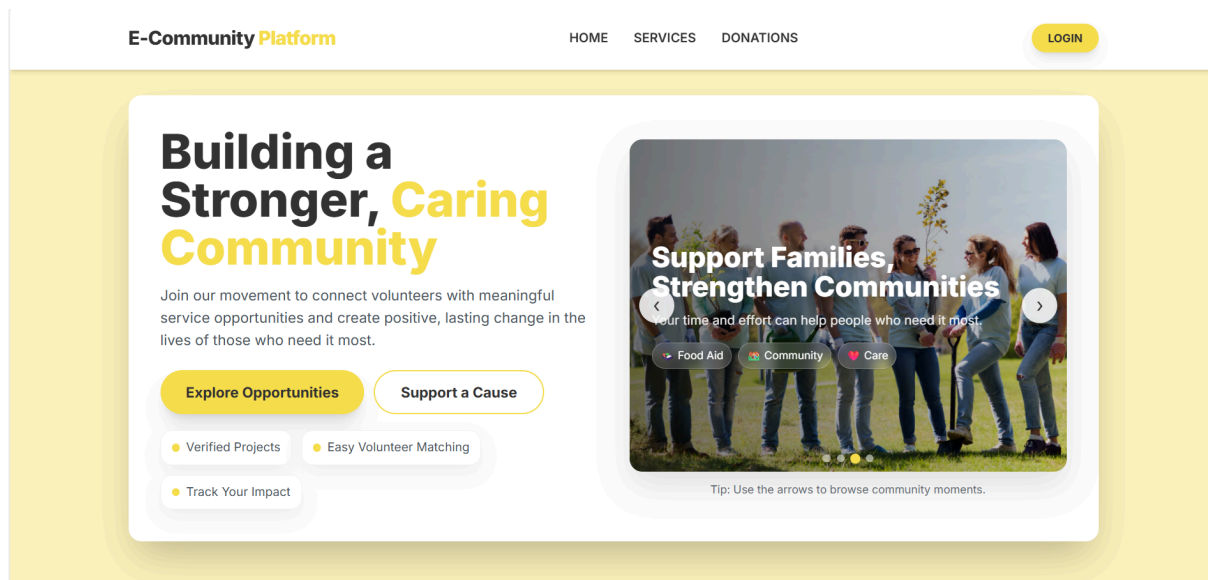


Figure 1.0 : Homepage of E-Community Platform

The “Explore Opportunities” button allows users to access the “Services” page, where users need to login first then they will see the available programs and volunteer activities are displayed. Users can browse and filter the list of events and view more details about each opportunity.

The “Support a Cause” button directs users to the “Support Our Cause” section on the dashboard. In this section, users can make donations to support community programs and help sustain the platform’s activities.

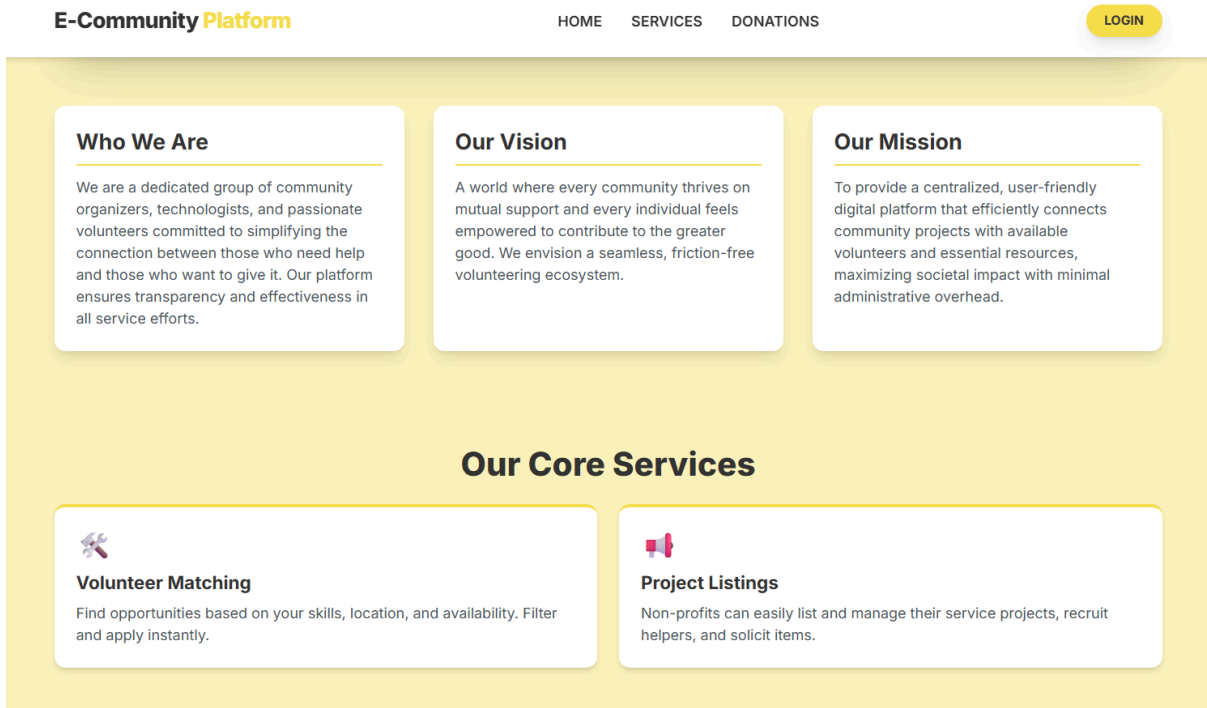


Figure 1.1 : Deep Insight Homepage of E-Community Platform

Users can find out the overview of the platform in this section of the dashboard.

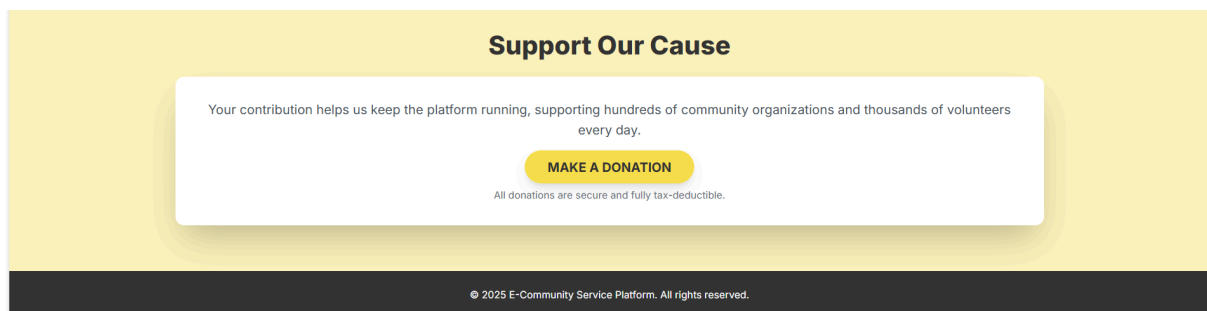


Figure 1.2 : Support Our Cause Homepage of E-Community Platform

From the dashboard, users can click the “Make a Donation” button. If the user is not logged in, the system will first prompt them to log in. After successful login, the user will be redirected back to the dashboard, where they can click the “Make a Donation” button again to proceed with making a donation.

2. Login Page

Figure 2.0 : Login Button at Dashboard of E-Community Platform

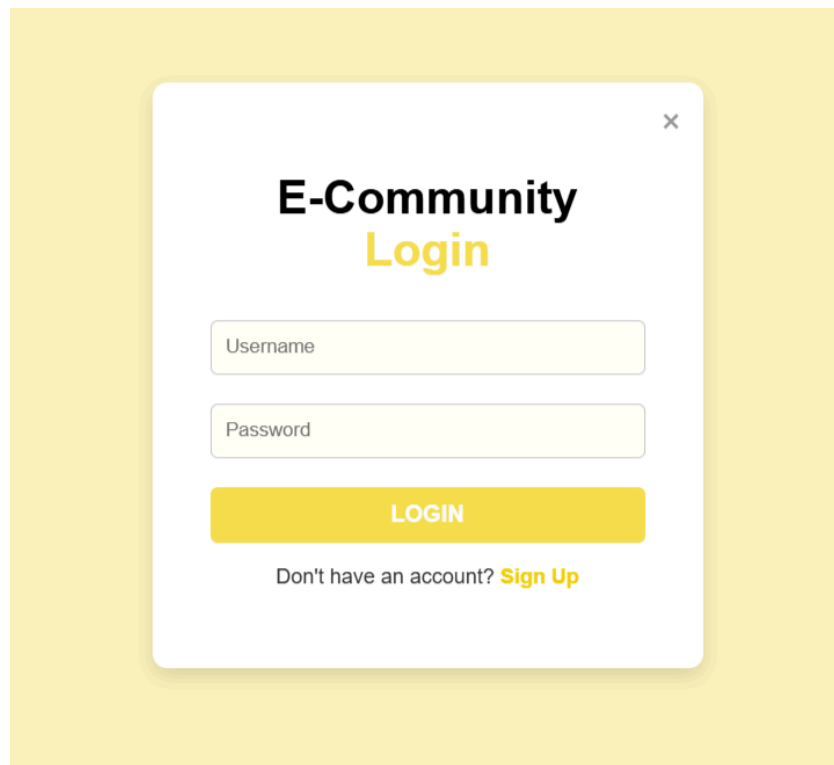
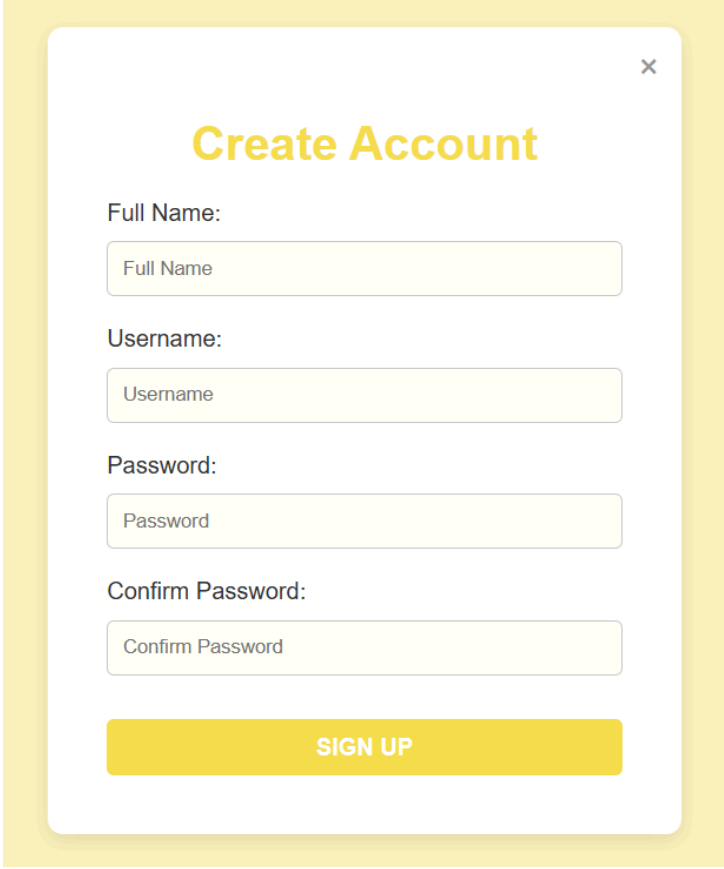
A login form titled "E-Community Login" is displayed on a yellow background. The form is a white card with a close button (X) in the top right corner. It contains two input fields: "Username" and "Password". Below these fields is a yellow "LOGIN" button. At the bottom of the form, there is a link that says "Don't have an account? Sign Up".

Figure 2.1 : Login Page of E-Community Platform

If a user clicks on the login button on the top right of the dashboard as shown in Figure 2.0, each user is required to enter their credentials before performing any actions like username and password in Figure 2.1. After a successful login, users are redirected to their respective pages. Regular users will be returned to the same page as shown in Figure 1.0 while administrators will be redirected to the designated admin page as shown in Figure 3.0. If the login was not successful, the login page will not continue the user request.

3. Sign-Up Page

A sign-up form titled "Create Account" with a close button (X) in the top right corner. The form contains four input fields: "Full Name", "Username", "Password", and "Confirm Password". Below the input fields is a yellow "SIGN UP" button. The form is set against a light yellow background.

×

Create Account

Full Name:

Username:

Password:

Confirm Password:

SIGN UP

Figure 3 : Sign Up Page of E-Community Platform

Before a user can log in, they must sign up if they do not have an existing account. During the registration process, users are required to enter their full name, username, password, and confirm their password to ensure that it matches. After clicking the Sign Up button, the user can proceed to log in to the system.

4. Admin Dashboard

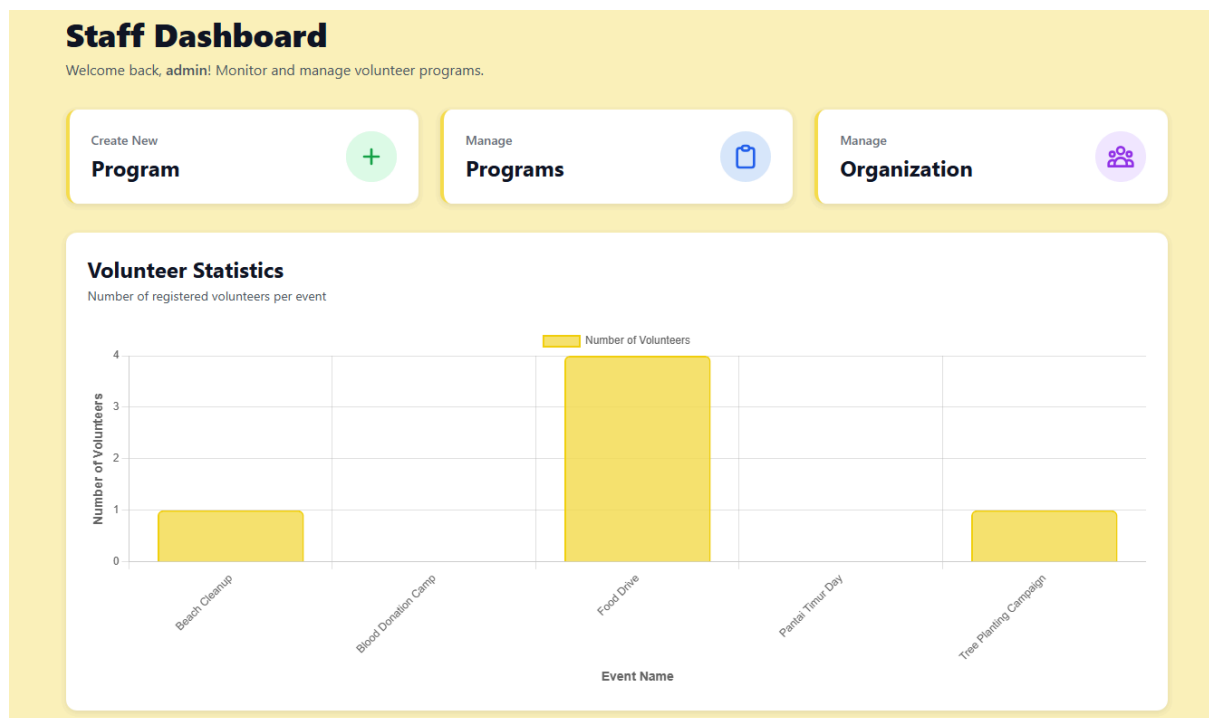


Figure 4.0. : Administrator Dashboard of E-Community Platform

After logging in as an admin, the administrator will see the dashboard which provides quick access to create new programs, manage programs and manage organizations. The dashboard also includes a volunteer statistics graph that displays the number of registered volunteers for each event.

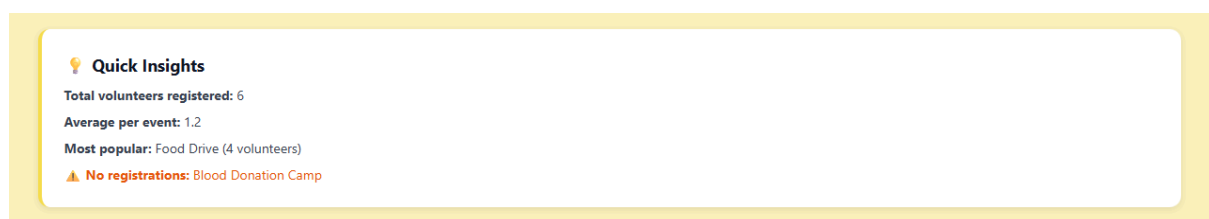


Figure 4.1 : Quick Insight of Administrator Dashboard

The Quick Insights section illustrated in Figure 4.1 provides a quick overview of volunteer participation on the platform. It shows the total number of registered volunteers, the average number of volunteers per event and the most popular event. It also highlights events with no registrations, helping users quickly identify areas that may need more attention.

5. Create New Program

The screenshot shows the 'Create New Program' form in the E-Community Platform. The form is centered on a yellow background. It includes the following fields and elements:

- Program Title ***: A text input field with the placeholder 'e.g., Beach Clean-Up'.
- Select Organization ***: A dropdown menu with the placeholder '-- Choose an Organization --'. Below it, a small text label reads 'The organization that will host this event'.
- Description ***: A text area with the placeholder 'Provide a brief description of the activities and goals.'.
- Date ***: A date picker field with the placeholder 'mm/dd/yyyy --:-- --'.
- Location ***: A text input field with the placeholder 'e.g., City Center, School Hall'.
- Buttons**: A 'Cancel' button and a 'Create Program' button at the bottom right.

Figure 5 : New Program Form Page of E-Community Platform

When the admin clicks on the Create New Program option, the system redirects the admin to the Create New Program page as shown in Figure 5.0. On this page, the admin is required to enter the program title, select an organization, provide a description of the organization and specify the date and location. If the Create Program button is clicked, the system will redirect the admin to the Program management dashboard page as shown in Figure 6.0. If the Cancel button is clicked, no data will be saved and the admin will also be redirected to the staff dashboard page shown in Figure 4.0.

6. Program Management Dashboard

The screenshot shows the 'Program Management Dashboard' in the E-Community Platform. The dashboard has a yellow background and includes the following elements:

- Header**: 'E-Community Platform' logo and 'Welcome, admin! LOGOUT' button.
- Left Sidebar**: A 'Dashboard' button.
- Main Content Area**:
 - Program Management Dashboard**: Title and subtitle 'Manage volunteer opportunities efficiently.'
 - Available Programs**: A section with a '+ Create New Program' button.
 - Program Cards**:
 - Beach Cleanup**: Location 'Pantai Cenang, Langkawi', Date '15-02-2026', Time '9:00 AM', and description 'Join us for a community beach cleanup event'. Includes 'Edit' and 'Delete' links.
 - Food Drive**: Location 'Community Center, Kuala Lumpur', Date '20-02-2024', Time '10:00 AM', and description 'Help collect food for families in need'.

Figure 6.0 : Program Management Dashboard of E-Community Platform

Selecting “Manage Program” on the dashboard illustrated in Figure 4.0 redirects the administrator to the view shown above in Figure 6.0. This page also appears by default after a new program is created. From this central list, administrators can manage the program by updating current program data or deleting programs that are no longer active respectively shown in Figure 6.1 and 6.2 below.

The screenshot shows a web interface for editing a program. The background is a solid yellow color. In the top left corner, there is a yellow button with a left-pointing arrow and the text "Back". Centered on the screen is a white form titled "Edit Program". The form contains several input fields: "Program Name" with the value "Blood Donation Camp", "Description" with the value "Donor registration and blood drive", "Location" with the value "Community Health Center", "Date & Time" with the value "01/25/2026 12:00 PM", and a dropdown menu for "Select Organization *" with the selected value "Community Helpers". Below the dropdown menu, there is a small text label: "The organization that will host this event". At the bottom of the form are two buttons: a grey "Cancel" button and a blue "Save Changes" button.

Figure 6.1 : Program Edit Form of E-Community Platform

The interface in Figure 6.0 includes an edit button allowing the administrator to modify key program details including the name, description of the event, location, date and time or associated organization as shown on Figure 6.1 above. Upon clicking Save Changes, the system commits these updates to the database and automatically refreshes the list displayed in Figure 6.0.

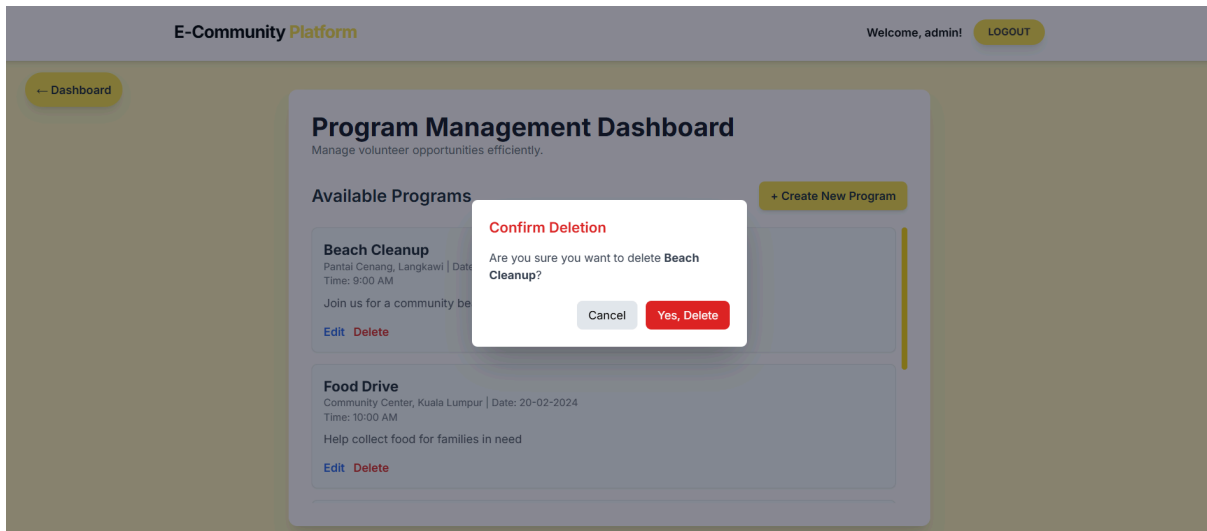


Figure 6.2 : Deletion Program of E-Community Platform

Furthermore, Figure 6.0 also has the delete button to remove the non active volunteer program to eliminate the unused data to maintain the database integrity. Upon clicking, a pop-up window will display the confirmation message to ensure the user really wants to delete it, if yes click on the red button “Yes delete” while otherwise click on the “Cancel” button.

7. Manage organization

Figure 7.0 below illustrates the organization list managed by the administrator. To facilitate effective communication, the system stores the name and contact email for each organizer. This ensures that users can easily locate contact details and reach out directly with questions or concerns regarding specific volunteer initiatives.

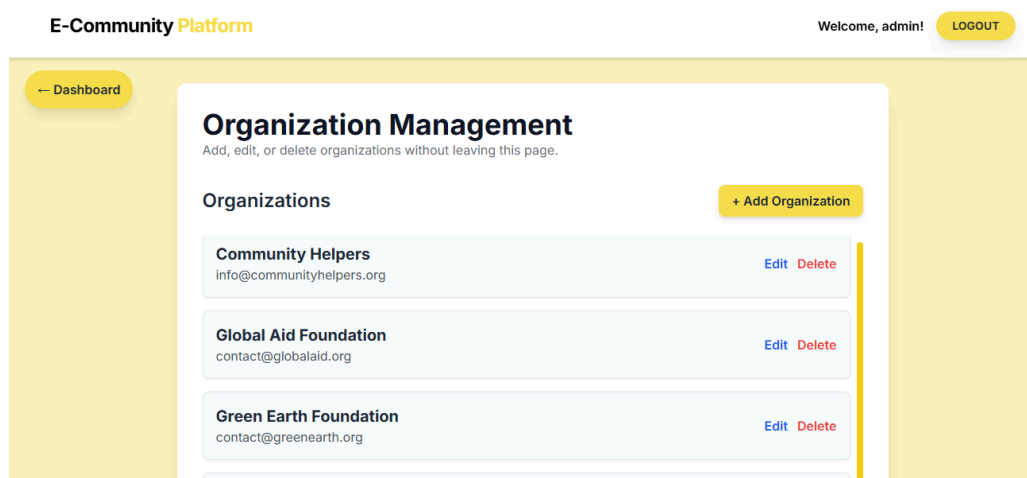


Figure 7.0 : Organization Management Page of E-Community Platform

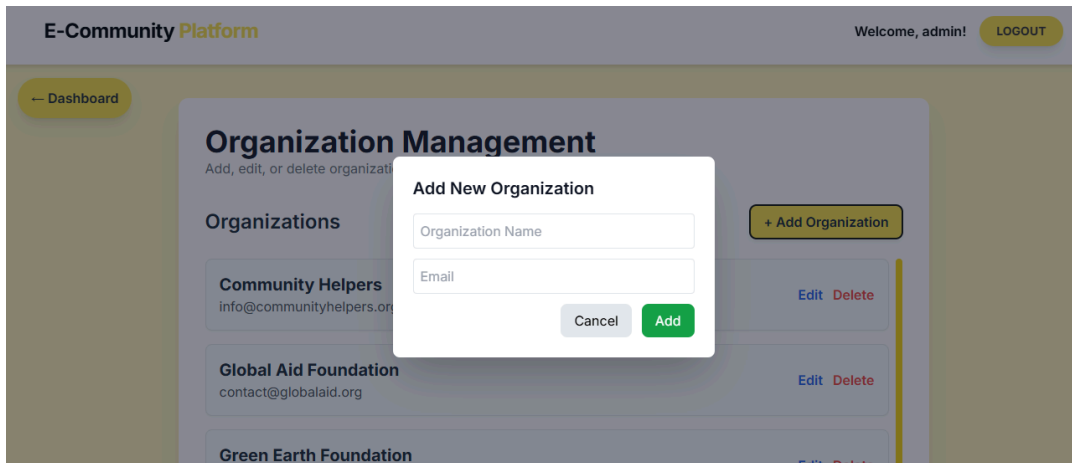


Figure 7.1 : Add Organization Page of E-Community Platform

As shown in Figure 7.1, administrators have a privilege to add a new organization who just participates to the system by entering the organization name and email of the organization. The successful message will pop-up once the system successfully stores the data.

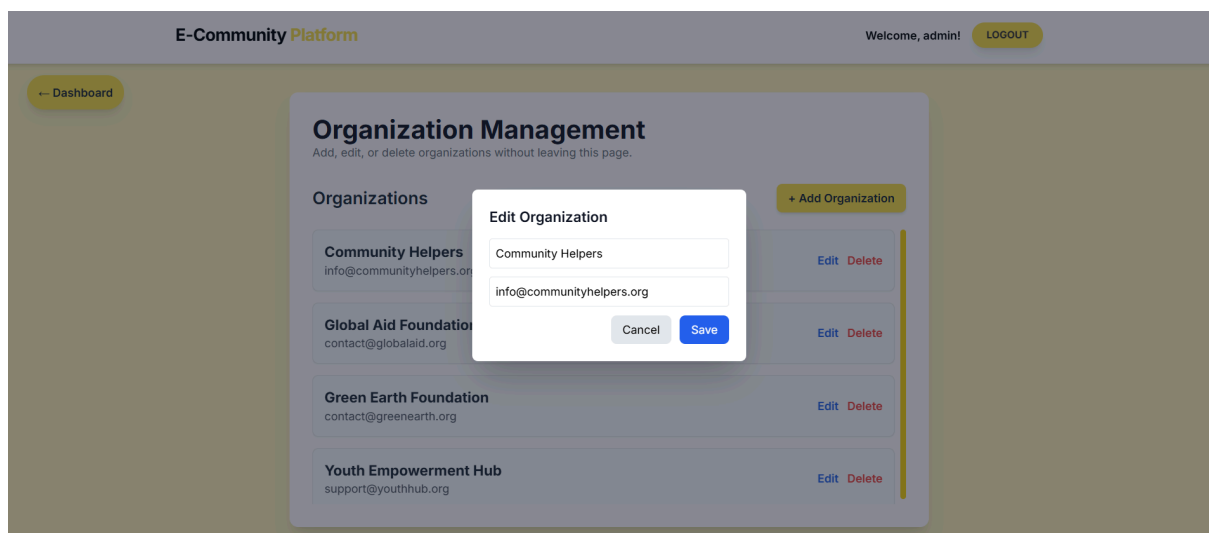


Figure 7.2 : Edit Organization Page of E-Community Platform

Figure 7.2 above illustrates that the admin also has the ability to edit existing records to fix the email or the name of the organization that is incorrect. Upon saving, the data is instantly updated in the database and on the screen. A confirmation message appears to verify the successful update.

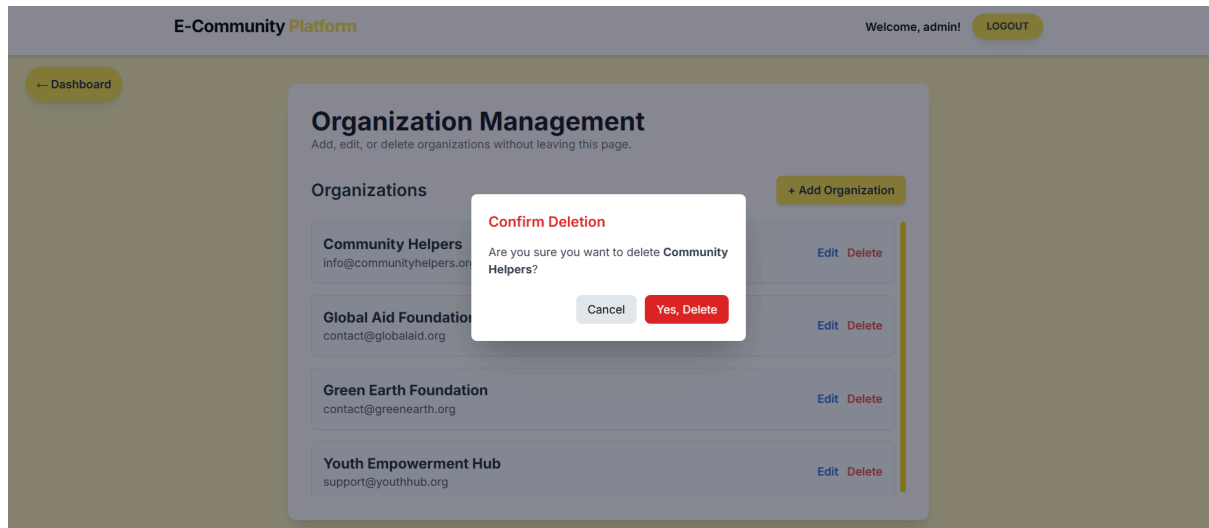


Figure 7.3 : Delete Organization Details of E-Community Platform

As shown in Figure 7.3, administrators have the option to delete an organization if it terminates its service with the platform. Upon the confirmation of deletion, the corresponding record is permanently removed from the database and the change is immediately reflected on the user interface. Lastly, it will show the success message of the deletion.

8. User Dashboard

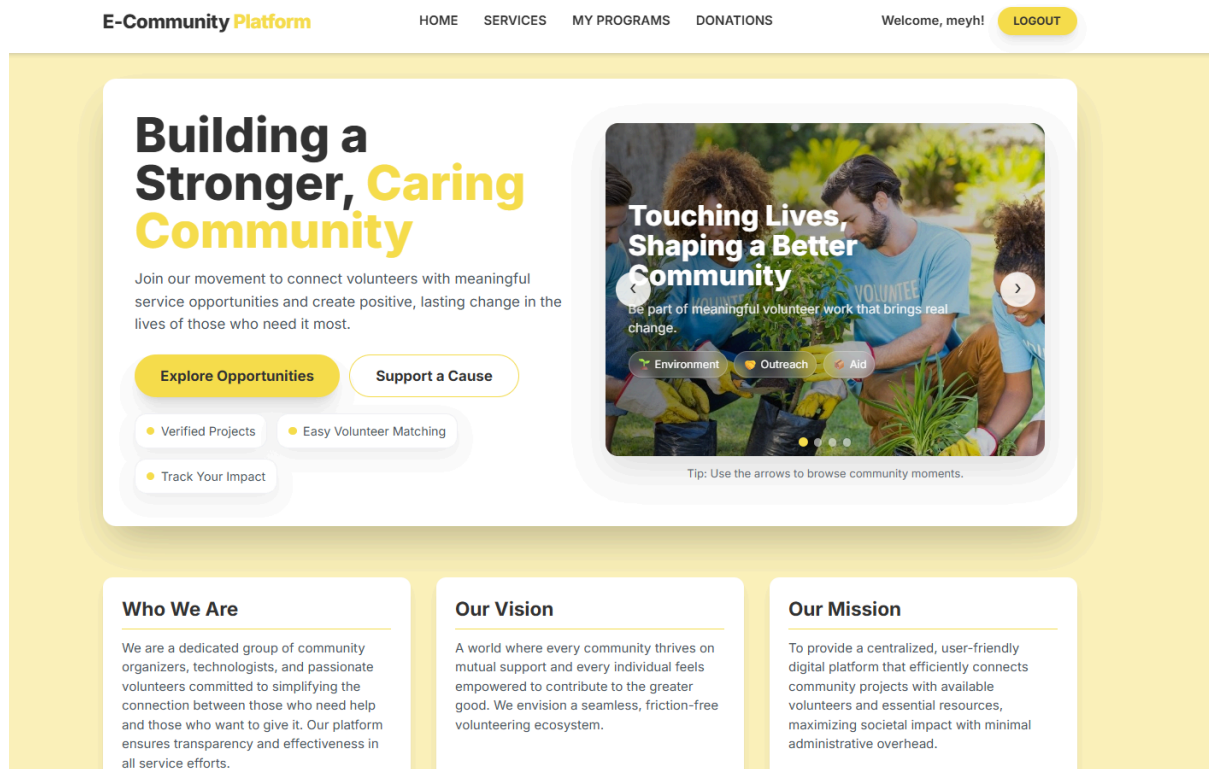


Figure 8 : The User Dashboard of E-Community Platform

The User Dashboard serves as the main landing page after a user successfully logs into the E-Community Platform. This page provides an overview of the platform's purpose and allows users to quickly access key features such as available programs, personal registrations, and donation options. From the dashboard, users can navigate using the top menu bar, which includes Home, Services, My Programs, and Donations.

9. Available Programs Page

E-Community Platform HOME SERVICES MY PROGRAMS DONATIONS Welcome, meyh! [LOGOUT](#)

Available Programs

Filter Events
Search Event: Location: Date From:

Blood Donation Camp
Description: Donor registration and blood drive
Location: Community Health Center
Date: 25-01-2026
Time: 12:00 PM
[Apply](#)

Pantai Timur Day
Description: asdada
Location: Pantai Teluk Kemang
Date: 31-01-2026
Time: 7:30 AM
[Apply](#)

Beach Cleanup
Description: Join us for a community beach cleanup event
Location: Pantai Cenang, Langkawi
Date: 15-02-2026
Time: 9:00 AM
[Apply](#)

Tree Planting Campaign
Description: Planting trees in community parks
Location: City Park
Date: 17-03-2026
Time: 8:30 AM
[Apply](#)

Figure 9.0 : List of the Available Programs Page

The Available Programs page displays a list of community service programs that users can participate in. Each program card provides essential information, including the program name, description, location, date, and time. Users may browse through the available programs and identify activities that match their interests. At the top of the page, users can filter programs using the Search Event, Location, and Date From filters. These filters help users quickly find specific programs based on their preferences.

The screenshot shows the 'Available Programs' page of the E-Community Platform. A modal form titled 'Apply for: Blood Donation Camp' is open, requiring the user to enter their Name, Phone Number, and IC Number. The background shows a list of programs, including Blood Donation Camp, Beach Cleanup, and Tree Planting Campaign, each with an 'Apply' button.

E-Community Platform HOME SERVICES MY PROGRAMS DONATIONS Welcome, meyh! LOGOUT

Available Programs

Filter Events

Search Event: Search by name... Location: All Locations Date From: mm/dd/yyyy

Blood Donation Camp
Description: Donor registration and blood donation event
Location: Community Health Center
Date: 25-01-2026
Time: 12:00 PM
Apply

Beach Cleanup
Description: Join us for a community beach cleanup event
Location: Pantai Cenang, Langkawi
Date: 15-02-2026
Time: 9:00 AM
Apply

Tree Planting Campaign
Description: Planting trees in community parks
Location: City Park
Date: 17-03-2026
Time: 8:30 AM
Apply

Apply for: Blood Donation Camp

Name..

Phone Number..

IC Number..

Cancel Submit

Figure 9.1 : Apply for Program Form Page of E-Community Platform

When a user clicks the Apply button on a selected program, the system displays a pop-up registration form as shown in Figure 9.1. This pop-up requires the user to enter their name, phone number, and IC number before submitting the application. After completing the required details and clicking the Submit button, the system processes the registration.

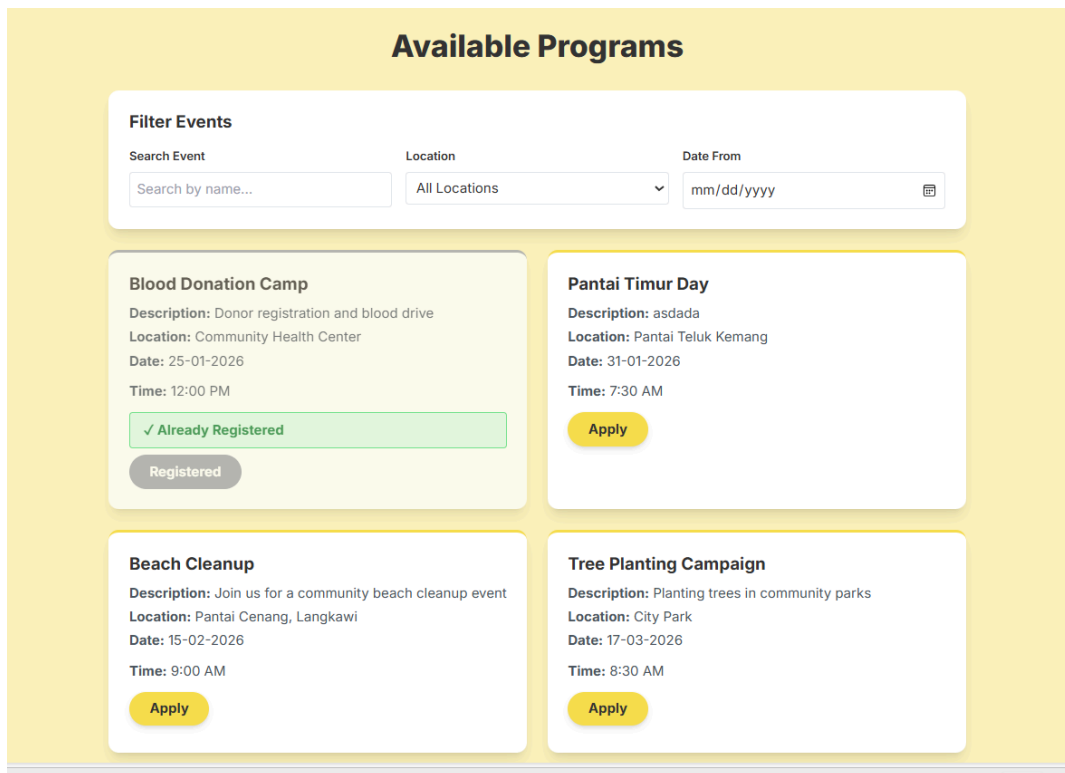


Figure 9.2 : Successful Registered Program of E-Community Platform

After the system completes the registration process, a confirmation message is displayed to inform the user that the registration has been successfully submitted. Once the registration is confirmed, the program card is automatically updated. The Apply button is disabled, and a green “Already Registered” status label is displayed. This visual indicator helps users easily identify programs they have already registered for and prevents duplicate registrations.

10. Program History Page

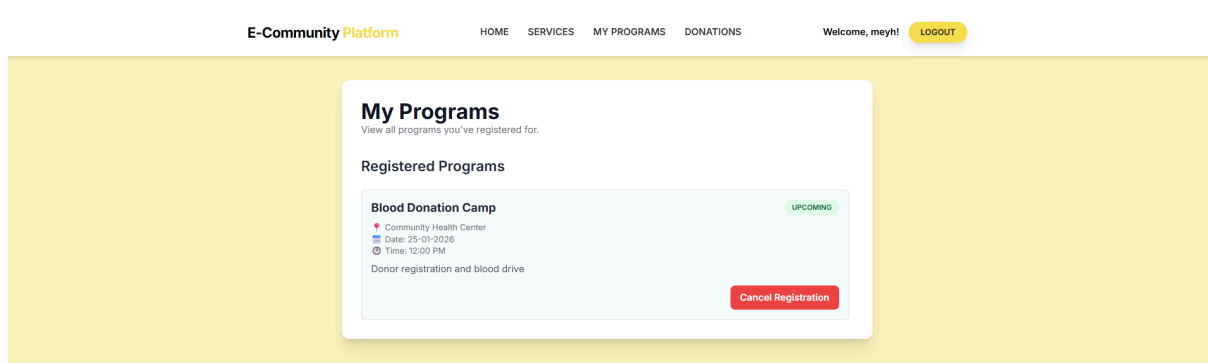


Figure 10 : History Joined Programs of E-Community Platform

The My Programs page as shown in Figure 10 allows users to view a list of programs they have registered for. This page helps users keep track of their active community service involvement and review program details such as dates, locations, and schedules.

11. Donations

Figure 11 : Donation page of E-Community Platform

In Figure 11, users can access the donation feature through the Donations menu or the Support a Cause button on the dashboard. If the user is not logged in, the system will prompt them to log in before proceeding. Upon successful login, the user is redirected back to the dashboard and can access the donation section to support community initiatives.

As illustrated in Figure 5.3, donors may choose from pre-set donation amounts or manually enter a preferred amount. Users are also provided with flexible payment options, including debit card, credit card, or QR code payment.